

Seed robustness of the learnable DCT-IV basis

100 random real-orthogonal initialisations, normal top-20% training (1008 steps), DIV2K-8q

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Question

Does normal training of the learnable DCT-IV basis depend on the random draw? `DCT4Basis(8,8)` is the real-orthogonal, ancilla-free analogue of the QFT: at its **canonical** init the forward operator is the exact bit-reversed orthonormal DCT-IV (30.54 dB @ $\rho\{=\}.20$ on the held-out test set). Following the QFT seed study’s init policy, we do **not** start there: each of 100 seeds s reseeds **everything trainable** — a fresh **Haar real-orthogonal** gate init (`dct4_random_basis(8,8,s)`: $\text{SO}(2)/\text{SO}(4)$ on the gates, a random sign on the controlled-phase gate, so the operator stays real-orthogonal), the 50-image training-batch subsample (from the fixed 500-image pool), and the training RNG — while the held-out **test set is held fixed** (canonical seed 42). Unlike the QFT writeup the operator is trained with **normal** batched training (`pdft.train_basis_batched`, one cosine-LR top-20% run of 1008 steps), not the gate-unfreeze sweep. Test PSNR is scored at four keep ratios ρ .

The random initialisations are genuinely different

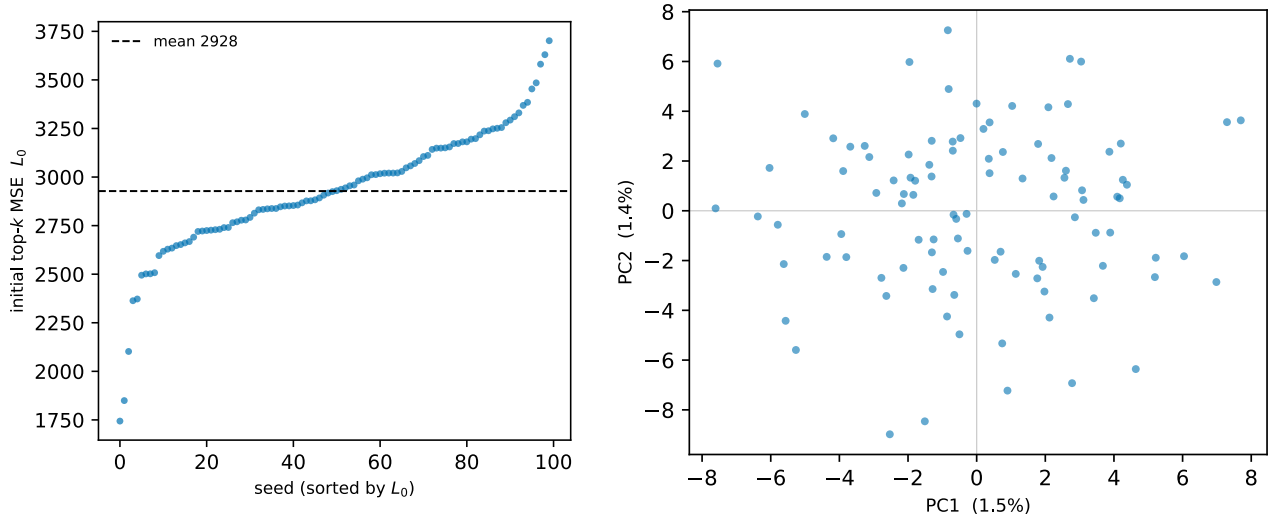


Figure 1: **(a)** the untrained random operator’s initial top- k MSE loss L_0 per seed — it spans 1743.9–3701.7 (mean 2927.8), so the seeds start from genuinely different, far-from-trained points. **(b)** a 2-D PCA of the 100 init parameter vectors; the inits are spread across many near-orthogonal directions rather than clustered.

Endpoint variance

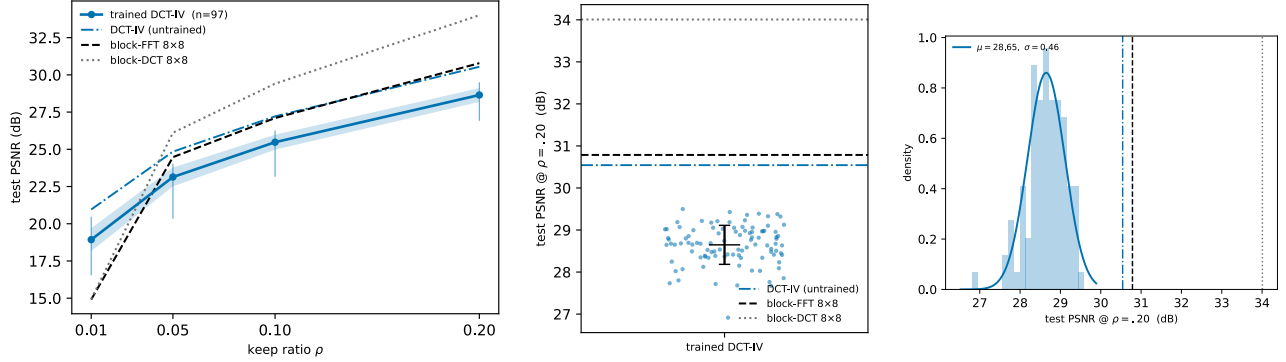


Figure 2: Test PSNR across 100 seeds. **(a)** mean $\pm\sigma$ band (+ min–max whiskers) vs ρ , against the untrained exact DCT-IV (dashdot), block-FFT 8×8 (dashed) and block-DCT 8×8 (dotted). **(b)** per-seed scatter at $\rho\{=\}.20$ with the mean $\pm\sigma$ bar. **(c)** the endpoint distribution at $\rho\{=\}.20$ (histogram + fitted normal); references marked.

basis	$\rho\{=\}.01$	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$	min@.20
trained DCT-IV	18.93 $\pm$ 0.77	23.14 $\pm$ 0.63	25.48 $\pm$ 0.5	28.65 $\pm$ 0.46	26.91
DCT-IV (untrained)	20.96	24.84	27.21	30.54	—
block-FFT 8×8	14.91	24.47	27.1	30.79	—
block-DCT 8×8	14.93	26.11	29.41	34.01	—

mean $\pm\sigma$ test PSNR (dB), $n = 100$ random-init seeds; min@.20 is the single worst seed at $\rho\{=\}.20$.

Reading

Despite starting from 100 genuinely different Haar real-orthogonal inits (far from trained — L_0 mean 2927.8), 1008 steps of normal top-20% training concentrate the endpoint into a narrow band: at $\rho\{=\}.20$ the per-seed standard deviation is 0.46 dB (mean 28.65, min 26.91, max 29.5). So the random draw causes only small turbulence in the endpoint.

Against the untrained exact DCT-IV (30.54 dB @ $\rho\{=\}.20$), 0 of 100 trained seeds match-or-exceed it; against block-FFT 8×8 (30.79 dB), 0 of 100 clear it. Block-DCT 8×8 (34.01 dB) remains the strongest classical transform here.

The per-seed endpoints' Shapiro–Wilk normality $p = 9.57 \times 10^{-3}$ is reported in `seed_sweep.json`; the fitted normal in the endpoint panel is a visual summary of location and spread.

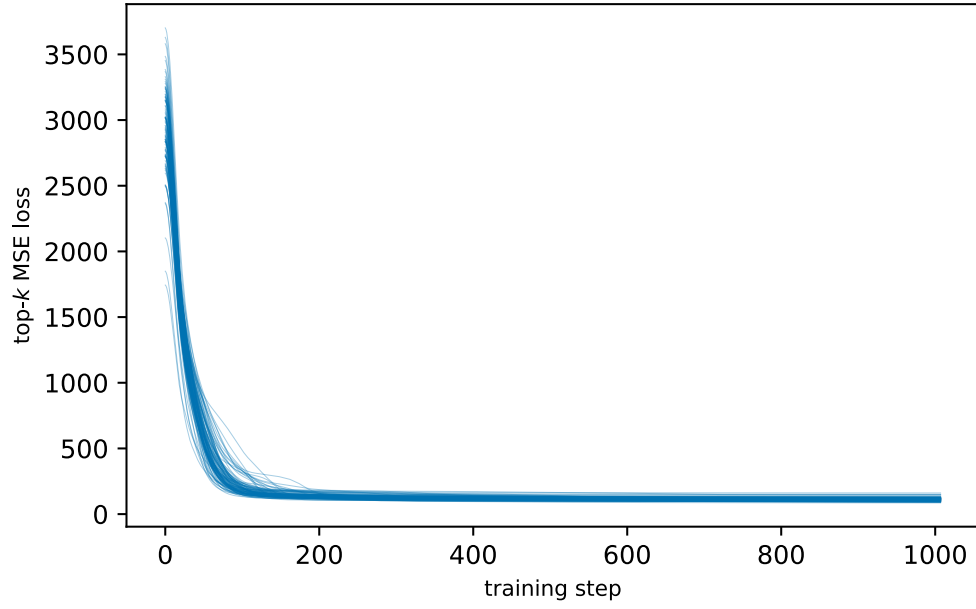


Figure 3: Per-seed training dynamics: every seed’s top- k MSE descent (one curve per seed) on a shared y-range. The Haar real-orthogonal starts descend from genuinely different, far-from-trained losses and converge into the narrow endpoint band above.